

Resolving the Two-Way Interspecies Communication Problem: An Intra-Call Acoustic-First Methodology

Introduction: The Convergence of Artificial Intelligence and Bioacoustics

The study of animal communication is currently undergoing a profound and irreversible methodological paradigm shift, catalyzed by the rapid convergence of large-scale bioacoustic data acquisition and unprecedented advances in artificial intelligence and machine learning.¹ Historically, ethological research and the analysis of animal vocalizations have been severely constrained by narrow, species-specific hypotheses, manual analytical techniques, and the immense human labor required to annotate comparatively small acoustic corpora.¹ For decades, researchers relied on human-synthesized sounds, such as acoustic lures designed primarily to increase trapping capture rates, rather than engaging in dynamic acoustic interactions.⁵ However, the widespread deployment of passive acoustic monitoring systems—ranging from terrestrial sensor networks in dense rainforests to deep-sea hydrophone arrays spanning entire ocean basins—has generated vast repositories of continuous environmental audio.⁴ One of the most prominent examples of this data explosion is the Orchiade, a massive repository acquired over twenty-five years via a network of hydrophones in northern British Columbia, containing countless hours of continuous marine mammal vocalizations.⁴

This immense influx of continuous, high-fidelity acoustic data has birthed the discipline of Animal Language Processing (ALP), a foundational scientific approach that utilizes large-scale artificial intelligence models to decode non-human communication at an unprecedented scale.¹ The emergence of digital bioacoustics relies on microscopic microphones and advanced bio-loggers that perpetually record sounds and bodily movements, allowing computational models—specifically Large Language Models (LLMs) and advanced neural architectures—to identify intricate patterns, hierarchical schemes, and latent syntactic structures within animal vocalizations.⁷ Organizations such as the Earth Species Project and the Cetacean Translation Initiative (CETI) are pioneering this frontier, developing sophisticated models trained on massive, diverse datasets spanning human speech, music, and environmental bioacoustics to learn foundational representations of bioacoustic signals across the tree of life.⁹ For instance, the NatureLM-audio model represents a pioneering large audio-language framework capable of zero-shot detection, classification, and unsupervised structural discovery across diverse and previously unencountered taxa.¹⁰

Despite these formidable analytical advances, the central, unresolved challenge within Animal Language Processing remains the "two-way interspecies communication problem".¹ Resolving

this problem requires systems capable not merely of passively parsing, segmenting, and classifying animal vocalizations, but of autonomously generating contextually appropriate, biologically accurate, and physically plausible acoustic responses in real time.⁷ True two-way communication mandates a closed-loop interactive paradigm wherein an artificial agent can perceive an incoming biological signal, rapidly map it to a specific behavioral or semantic context, formulate a strategic reply based on interaction probabilities, and synthesize the acoustic output within the natural temporal rhythms and acoustic parameters of the target species.¹³ Achieving this represents the ultimate realization of a biological Turing test.¹⁵

To engineer such a system, traditional anthropocentric linguistic frameworks—which often inappropriately superimpose human syntactic rules, lexical categories, and phonological assumptions onto non-human signals—must be entirely discarded in favor of an "acoustic-first" methodology.¹⁷ This approach treats the raw acoustic physics of the signal, including its continuous micro-dynamics and multidimensional texture, as the primary substrate of meaning.⁴ The following report exhaustively details a comprehensive scientific methodology and operational protocol for solving the two-way interspecies communication problem. It outlines the requisite data hierarchies, the mathematical architecture of neural boundary detection and continuous feature extraction pipelines, the real-time closed-loop agent protocols, and the biologically grounded bioacoustic synthesis engines. Furthermore, it analyzes critical empirical findings in highly vocal mammalian models, demonstrating how these advanced technologies unearth combinatorial syntax, crowd-based vocal learning, and profound semantic specificity, ultimately laying the rigorous methodological groundwork for coherent interspecies dialogue.

The Paradigm Shift: From Anthropocentric Linguistics to Acoustic-First Methodologies

For nearly a century, the study of animal communication has been heavily biased by the frameworks of human linguistics. Researchers historically sought direct analogs to human phonemes, words, and discrete syntactic structures within the vocalizations of birds, marine mammals, and terrestrial quadrupeds.¹⁸ However, attempting to map animal calls to human linguistic architecture is a fundamentally flawed premise. Human speech production is heavily reliant on highly specific anatomical mechanisms—such as the complex kinematics of jaw height, tongue placement, and the resulting manipulation of the first and second acoustic formants.¹⁷ Studies utilizing predictable and unexpected auditory feedback perturbations in humans demonstrate that our sensorimotor control of speech is intrinsically linked to these specific articulatory kinematics, which are largely absent or radically different in other species.¹⁷

Consequently, traditional bioacoustics frequently resorted to analyzing easily measurable "inter-call" variables, such as the macro-sequencing of discrete barks, clicks, or whistles, or the temporal duration between distinct vocal events (inter-call intervals).²² While analyzing inter-call intervals has proven valuable for measuring basic turn-taking latency, such as the temporal distribution of antiphonal phee calls in marmosets²⁷ or the sequence ordering in chickadee alarm calls⁵, macro-level inter-call analysis systematically ignores the immense informational

density contained continuously within the acoustic envelope of a single vocalization.⁴

An "intra-call" acoustic-first methodology recognizes that the subtle, continuous modulations within a single vocal boundary—such as micro-fluctuations in fundamental frequency (F0), rapid shifts in spectral energy distribution, and dynamic phase perturbations—serve as the primary biological carriers of emotive and semantic data.⁴ For example, detailed behavioral analyses of avian brood parasites, such as common cuckoos, demonstrate that crucial information is encoded in intra-call variations, specifically the precise temporal pause between adjacent

syllables and the subtle difference in maximum frequencies (ΔF_{max}) between those syllables.³⁰ Similarly, mammalian communication is rich with intra-call continuous variation. Research indicates that the purrs and grunts of female mammals shift continuously from tonal elements to noisy, harsh acoustics depending strictly on proximity to a threat and internal affective states.³¹ Brain stimulation studies in squirrel monkeys further confirm this biological mapping, demonstrating that harsh, low-frequency acoustic textures inherently correspond to aversive emotional states, whereas high-frequency tonal sounds correlate with positive, self-stimulating states.³¹ Furthermore, aggressive close-approach howling between wolf packs utilizes significantly deeper pitch within the intra-call structure compared to howling conducted from greater distances.³¹

These phenomena highlight the imperative requirement to abandon discrete human phonetic labeling and adopt a methodology that models the vocalization purely through its acoustic physics and prosodic texture.⁴ The acoustic-first methodology operationalizes this by prioritizing the physics of the direct sound wave—the "acoustic first impression"—over arbitrary semantic categorization.¹⁷ It extracts high-dimensional, continuous feature vectors that capture the precise physiological state of the animal's vocal tract and the resulting acoustic geometry, allowing machine learning models to discover latent linguistic structures organically without human prejudice.¹³

Dataset Requirements: The Data Spectrum Hierarchy

A critical computational bottleneck in modern bioacoustics is the severe disparity between continuous, unstructured raw field recordings and the heavily segmented, meticulously annotated datasets traditionally required for supervised machine learning pipelines.³² Standard models frequently demand that audio be pre-segmented into discrete clips prior to ingestion.³³ This assumption catastrophically fails when analyzing the graded, continuously overlapping vocal streams typical of complex, dense animal societies.³³

To overcome this fundamental limitation, modern acoustic-first architectures utilize Neural Boundary Detection (NBD) algorithms capable of autonomously extracting sequential acoustic gestures directly from raw, unsegmented hydrophone or microphone dumps.⁴ Consequently, the primary bottleneck for establishing an interspecies communication project is no longer the manual segmentation of the signal, but rather the accurate, time-aligned annotation of behavioral and social context.¹³ To formalize the data collection process, the methodology

establishes a rigorous four-tiered "Data Spectrum" hierarchy, categorizing the utility of bioacoustic datasets for specific machine learning and interactive synthesis tasks.

Data Level	Nomenclature	Core Compositiona l Requirements	Engine Capabilities and System Processing	Analytical Outcomes and Inherent Limitations
Level 0	Structural Discovery	Continuous, raw acoustic recordings. Zero human-applied labels or manual temporal segmentation required.	Neural Boundary Detection (NBD) autonomously isolates atomic units. Unsupervised syntactic mining extracts repeating N-gram templates.	System successfully delineates whether a species utilizes a Graded Continuum or Discrete Syntax. Cannot deduce semantic meaning due to an absolute lack of behavioral context mapping. ¹³
Level 1	Semantic Discovery	Raw acoustic recordings paired with synchronized, time-aligned Context Labels (e.g., "Traveling", "Feeding", "Alarm").	NBD creates the temporal sequence while supervised models map those sequential vectors to the provided behavioral contexts.	System successfully identifies meaning by statistically predicting behavioral context from acoustic syntax. Allows legacy isolated call datasets to be concatenated into

				"pseudo-sequences" as an upgrade path. ¹³
Level 2	Functional	Raw acoustic recordings + Context Labels + Emitter ID (Definitive identification of the calling individual).	System actively builds Individual Voice Libraries based on micro-acoustic profiling and clustering algorithms.	Successfully differentiates between "Shared Syntax" (universal, species-wide calls) and localized "Dialects" (specific to colonies or individuals). Crucial for identity-dominant taxa like bats. ¹³
Level 3	Interaction-Ready	Raw acoustic recordings + Context Labels + Emitter ID + Receiver ID (Definitive identification of the specific addressee).	System models comprehensive social dynamics, complex Hub-and-Spoke topologies, turn-taking latency, and directed prosody.	The absolute baseline requirement for deploying a two-way closed-loop agent. The system must possess the algorithmic capacity to decide who to reply to within a multiparty biological network. ¹³

The architectural progression from Level 0 to Level 3 represents the complex computational transition from mere unsupervised pattern recognition to actionable, context-aware biological interaction.¹³ While massive foundation models trained via self-supervised representation

learning can execute impressive zero-shot classification and unsupervised structural discovery on Level 0 data, allowing researchers to analyze massive ethology datasets in minutes rather than months³, deploying a responsive closed-loop agent inherently demands Level 3 data. Without explicitly identifying both the specific emitter and the intended receiver within a social network, generative algorithms cannot accurately formulate contextually appropriate interactive responses, risking the generation of nonsensical or biologically disruptive acoustic stimuli.¹³

Stage 1: Neural Boundary Detection and Semantic Segmentation

The first operational stage in the intra-call acoustic-first pipeline requires the precise, autonomous segmentation of continuous audio streams into discrete, semantically meaningful phrase units.¹³ Older computational bioacoustic pipelines relied almost exclusively on standard energy-based Change Point Detection (CPD) or simple amplitude thresholding to slice audio into manageable clips.¹³ However, CPD algorithms suffer from severe structural weaknesses when deployed against field recordings: they frequently fragment graded signals (where an animal shifts semantic meaning without pausing for breath) and routinely miss low-energy boundaries obscured by environmental noise, such as the subtle frequency modulations of ultrasonic bat sweeps.¹³

While the field has recently developed highly advanced deep learning solutions for segmentation—such as DeepSqueak, which utilizes cutting-edge regional convolutional neural network architecture (Faster-RCNN) to perform human-quality syllable classification in rodents³², or TweetyNet, a single neural network incorporating Long Short-Term Memory (LSTM) models to segment spectrograms of birdsong without clean audio prerequisites³³—these massive models often introduce latency and computational overhead that is strictly prohibitive for microsecond-reaction closed-loop interactive systems. Furthermore, some models rely exclusively on convolutional feature extraction without exploiting the sequential, temporal structure inherent to vocalizations, though hybrid tools combining CNNs and RNNs are beginning to close this gap.³²

The optimized Neural Boundary Detector (NBD) utilized in the current methodology resolves the latency and fragmentation issues by employing a temporal-convolution-inspired mathematical approach designed explicitly to detect semantic boundaries rather than mere energy gaps.¹³ Operating with a processing latency of less than one single frame and maintaining a

computational complexity of $\mathcal{O}(n)$ per audio sample, the NBD provides real-time capability essential for agent interaction.[13, 13] The detection process is executed through five primary algorithmic steps:

First, the system computes the Root Mean Square (RMS) energy profile for each highly localized temporal frame, subsequently normalizing these values against the global track to create a baseline amplitude envelope.¹³ Second, to detect timbral and semantic changes

entirely independently of overall energy levels—vital for recognizing a soft boundary where an animal seamlessly morphs from a tonal whistle to a broadband burst—the algorithm computes a Spectral Change Profile. This is achieved by calculating a spectral centroid using a rapid zero-crossing approximation, extracting the mathematical derivative representing the change from the preceding frame, and normalizing the result.¹³

Third, the NBD executes a Boundary Probability Fusion step. The independent energy and spectral features are mathematically combined using dynamically learned, species-specific weights to determine a final, continuous boundary probability score. The operating formula is defined as:

$$P_{boundary} = (w_{energy} \times \Delta E) + (w_{spectral} \times \Delta S)$$

where w_{energy} represents the learned weight for amplitude shifts, ΔE represents the normalized temporal energy change, $w_{spectral}$ is the weight for frequency shifts, and ΔS is the normalized spectral derivative.¹³

Fourth, the algorithm applies a Temporal Smoothing filter. A moving average is calculated over a specified number of smoothing_frames to suppress high-frequency environmental noise and prevent algorithmic false positives.¹³ Finally, a strict Debounce Logic is enforced. To prevent "rapid-fire" micro-segmentation caused by rapid vibrato or reverberation, the system enforces a minimum phrase duration (typically defaulting to 50 milliseconds). A calculated boundary is only formally accepted and registered into the system if the temporal distance since the last confirmed boundary strictly exceeds this minimum sample threshold.¹³

Crucially, unlike binary CPD systems, the NBD outputs highly descriptive NbdPhraseBoundary objects containing the temporal position, a continuous confidence score ranging from zero to one, and a specific phenomenological classification of the boundary type.[13, 13] These types include Hard boundaries (characterized by a definitive, measurable drop in acoustic energy to ambient silence), Soft boundaries (a semantic or spectral shift that occurs continuously without any drop in energy), and Transitional boundaries (a gradual, morphing change occurring smoothly over time).[13, 13] This segmented output, alongside its classification data, is then seamlessly fed into the next stage for high-dimensional feature extraction.

Stage 2: 112-Dimensional Rosetta Feature Extraction

Once the Neural Boundary Detector isolates a continuous acoustic segment, the system must project the raw waveform into a highly descriptive mathematical space. The acoustic-first methodology accomplishes this through the extraction of a 112-dimensional continuous feature vector, formally referred to as Rosetta Features.¹³ This extreme high-dimensional embedding deliberately eliminates the necessity for human-labeled symbolic categories, instead representing the animal vocalization purely through its foundational acoustic physics, nonlinear dynamics, and complex prosodic textures.¹³ By generating a continuous flattened array of 112

32-bit floats ([f32; 112]), the system creates an optimal structure for rapid machine learning ingestion and unsupervised clustering.¹³

The 112 dimensions are hierarchically organized into three distinct layers, designed to capture both the macro-level structural geometry and the minute intra-call variations that serve as individual identifiers and emotive carriers.¹³

Layer 1: Base Physics (46 Dimensions)

The foundational layer establishes the core acoustic geometry and absolute physical measurements of the biological signal.¹³ It maps the fundamental frequency (F0) parameters extensively, calculating the mean F0, the maximum and minimum frequency range utilized within the segment, and the overall trajectory of F0 drift over time. It extracts the absolute energy profile, tracking both the continuous RMS power and peak amplitude limits.¹³ To model the temporal amplitude envelope—which is strictly tied to the biomechanics of breath control and vocal fold vibration—the system calculates biological Attack-Decay-Sustain-Release (ADSR) dynamics, identifying the exact millisecond attack times and logarithmic decay phases.[13, 13] Furthermore, Layer 1 maps the overall spectral shape by quantifying the spectral centroid, spectral flatness, and spectral contrast.¹³ Importantly, it also isolates the Harmonic-to-Noise Ratio (HNR), providing an immediate mathematical indication of whether the biological signal is highly tonal and controlled, or chaotic and breathy, which often correlates directly with arousal or aggression.[13, 13]

Layer 2: Macro Texture (30 Dimensions)

Layer 2 is engineered to model the prosodic, rhythmic, and articulatory superstructure of the acoustic phrase.¹³ To approximate the resonant filter properties of the animal's vocal tract—analogueous to how formants shape human vowels—the system computes 13 Mel-Frequency Cepstral Coefficients (MFCCs) alongside 6 Delta MFCCs, which capture the temporal rate of change within the vocal tract geometry.¹³ Rhythmic features are heavily analyzed, recording the acoustic onset rate and the median inter-click intervals, which are particularly vital for marine mammals utilizing echolocation trains.¹³ Layer 2 also maps geometric approximations of the F0 contour, capturing the mathematical slope and curvature of the pitch trajectory. Detailed Frequency Modulation (FM) statistics are extracted to map large-scale sweeping vocalizations, an absolute necessity for profiling species that utilize rapid frequency sweeps for ranging or long-distance contact calling.¹³

Layer 3: Micro Texture (36 Dimensions)

The deepest and most complex layer of the feature extraction pipeline analyzes the micro-acoustic phenomena that frequently function as unique individual identifiers or convey acute, subtle shifts in emotional urgency.¹³ This layer captures fine-grained textural "fingerprints" by evaluating the spectral geometry through statistical skewness and kurtosis, measuring the asymmetry and tailedness of the frequency distribution.¹³ Crucially, the system borrows techniques from advanced image processing by computing Gray-Level Co-occurrence Matrix

(GLCM) Texture features directly upon the high-resolution spectrograms. This allows the system to quantify complex 2D spectral contrast, entropy, and local energy distribution, providing a mathematical representation of acoustic "roughness" or "smoothness".¹³ Layer 3 also evaluates micro-dynamics, specifically isolating the precise mathematical depth and cycle rate of vibrato (continuous pitch perturbation) and tremolo (continuous amplitude perturbation).¹³ Finally, it measures jitter (cycle-to-cycle phase perturbation) and shimmer (cycle-to-cycle amplitude perturbation), critical metrics traditionally associated with biological voice quality and physiological stress.[13, 13]

Stage 3: Corpus Analysis, Syntactic Mining, and the Lexical Vocabulary

To bridge the continuous, highly dimensional acoustic data with a discrete generative synthesis model, the extracted vectors must be statistically compressed into a manipulatable, finite vocabulary. The methodology accomplishes this through an offline "Teacher" distillation pipeline, executing advanced Python-based unsupervised clustering to discover the true statistical structure—or "Dense Acoustic Continent"—of a species' vocalizations.¹³

Processing massive datasets (e.g., up to 8.9 million segments), the Teacher pipeline first subsamples the data to make Expectation-Maximization (EM) training tractable without losing statistical validity.¹³ The 112D feature vectors are subjected to Principal Component Analysis (PCA), reducing the dimensionality to 30D while preserving 95.4% of the acoustic variance to accelerate computation.¹³ Unlike older algorithms such as MiniBatchKMeans or hard-boundary methods like HDBSCAN, the updated pipeline utilizes Bayesian Gaussian Mixture Models (BGMM).¹³ This approach allows the system to discover the natural, optimal number of clusters (e.g., successfully isolating 45 true semantic clusters in specific model organisms) while preserving continuous acoustic space.¹³

To further refine the lexicon, weight-based pruning eliminates anomalous clusters representing less than 1% of the dataset.¹³ The surviving mathematical centroids are then projected back into the full 112D space as "acoustic archetypes."¹³ The specific biological segment exhibiting the absolute minimum mathematical distance to these archetypes is preserved as the "exemplar."

This analytical stage produces a highly optimized `synthesis_manifest.json`.¹³ Each archetype is mapped to behavioral contexts (e.g., "alarm," "social," "contact") based strictly on its cluster identity, wrapped in a `SourceMetadata112D` structure, and mathematically scored based on RMS energy, HNR, jitter, and shimmer metrics to ensure that the synthesis engine only utilizes clean, unambiguous sounds for reproduction.¹³ These isolated exemplars and their metadata constitute the foundational acoustic alphabet with which the interactive closed-loop agent can begin to compose novel, contextually appropriate sentences.

Stage 4: The Closed-Loop Agent Protocol for

Interactive Playback

Historically, playback experiments in animal communication were strictly "open-loop." Researchers would record a vocalization, determine a hypothesis, and play the static recording back to the animal at a later time, observing the behavioral response.¹⁴ While useful for basic biological inquiries, open-loop systems completely strip away the dynamic temporal interplay, turn-taking latency, and context-dependent syntax modulation that define genuine two-way communication.⁵ A static, pre-recorded mapping of calls to contexts is entirely insufficient to solve the two-way interspecies communication problem; the algorithm must continuously update its sensory-motor maps and modulate its output to match the immediate, fluctuating affective state of the interspecies dialogue.¹⁴

To facilitate true interactive playback, the methodology employs a highly sophisticated, dual-layer closed-loop computing architecture, structurally bifurcated into a Rust-based Execution Layer and a Python-based Logic Layer.¹³ This intentional architectural division ensures that the high-speed, safety-critical digital signal processing and audio synthesis operations are strictly decoupled from the computationally heavy, non-deterministic cognitive inference models.¹³

To process these signals, the Rust Execution Layer functions as a high-performance "Student" capable of sub-millisecond real-time inference.¹³ It loads the `synthesis_manifest.json` via zero-copy deserialization into a `HashMap` for $O(1)$ lookups.¹³ When an incoming sound is detected, the Student performs a linear scan using L2 Squared distance to find the nearest centroid. Because the optimized dataset (e.g., 45 clusters at ~20 KB) fits entirely within the CPU's L1 cache, this linear scan actually outperforms complex hierarchical indices like HNSW, maintaining a massive throughput of up to 228,000 segments per second.¹³

ZeroMQ Transport and Inter-Process Communication

Given the extreme demands of bioacoustic processing—where interacting with ultrasonic species like bats requires real-time synthesis at sampling rates of 96 kHz—the communication link between the analytical Python node and the generative Rust node must operate with microsecond latency.^[13, 13] The protocol achieves this by utilizing ZeroMQ (v4.3+) executing exclusively over Unix domain sockets (IPC).¹³ This creates a zero-copy data transfer mechanism, bypassing the operating system's network stack entirely. By serializing all data into flat NumPy arrays, the system fundamentally eliminates the heavy processing overhead associated with JSON or XML parsing, ensuring uninterrupted real-time performance.¹³

The closed-loop architecture operates across three dedicated IPC channels:

1. **The Feature Channel (`ipc:///tmp/cognitive_features.ipc`):** The Rust Execution layer utilizes a `FeatureEventPublisher` to BIND and continuously stream the live, real-time 112D feature vectors and matched cluster IDs to the Python Logic Layer. To prevent algorithmic backpressure and buffering delays, this socket is configured with a High Water Mark

(SNDHWM) of 100 and exclusively utilizes non-blocking sends via the DONTWAIT flag.¹³

2. **The Action Channel (ipc:///tmp/cognitive_actions.ipc):** The Python agent computes the semantic response and utilizes an ActionPublisher to transmit a generative SynthesisAction payload back to the Rust ActionSubscriber for immediate acoustic execution.¹³
3. **The Heartbeat Channel (ipc:///tmp/cognitive_heartbeat.ipc):** This serves as the system's primary bioethical fail-safe mechanism. The Rust PeerController continuously monitors a steady heartbeat signal from the Python layer. If the cognitive logic layer stalls, panics, or crashes, the Rust layer immediately detects the disruption and falls back to a "Passthrough Mode." This instantly halts all generative audio output, preventing the system from broadcasting chaotic, out-of-context computational noise that could distress the interacting animal.¹³

The Four-Stage Cognitive Cycle

Within the Python Logic Layer, the advanced InteractionAgent (v1.2.0) operates continuously across a highly optimized, biological four-stage decision cycle: Perceive, Understand, Decide, and Act.¹³

In the **Perceive** stage, the agent's FeatureSubscriber receives the incoming sequence of 112D Rosetta Features, tracking the N-gram syntactical sequences generated by the live animal.¹³

During the **Understand** stage, the agent applies Cluster-Based Semantic Grounding. It uses the pre-computed cluster mappings to definitively assign a behavioral context to the incoming signal.¹³ Crucially, the Rust Student simultaneously enforces a Safety-Critical Perception Filter: if an incoming signal's mathematical distance to the nearest centroid exceeds a strictly set threshold, it is actively rejected as Out-Of-Distribution (OOD). This fundamentally prevents the agent from reacting to environmental noise or hallucinating a response to its own synthesized acoustic output.¹³

The **Decide** stage is governed by strict algorithmic constraints executed by the v1.2.0 agent.¹³

First, the agent calculates a confidence score based on the distance algorithm ($1.0 - \text{distance}/\text{threshold}$). If this confidence falls below a strict minimum (e.g., 0.5), Confidence-Based Suppression actively halts the response.¹³ Furthermore, the agent operates as a probabilistic automaton (e.g., a 45-state automaton mapping to the 45 discovered true clusters) to execute strict Syntax Validation.¹³ It checks the proposed interaction against a matrix of valid biological bigrams; a response is only synthesized if the transition between the last heard cluster and the proposed reply constitutes a biologically observed sequence, decisively preventing the generation of "acoustic gibberish."¹³

Finally, in the **Act** stage, the agent compiles a detailed SynthesisTimeline. This data structure packs the intended sequence of cluster IDs, the precise relative start times (in milliseconds), the required duration boundaries, and vital parameters for the Differentiable Digital Signal Processing (DDSP) pipeline.¹³ The system recently transitioned from limited 7D

MicroDynamicsDelta transformations to a full 112D DDSF architecture, enabling highly fine-grained generative control over emotional prosody with sub-50ms latency before it is transmitted back to the execution layer.¹³

Stage 5: Biologically Grounded Acoustic Synthesis

The terminal node of the acoustic-first methodology involves transforming the cognitive timeline back into physical acoustic waves that are biologically plausible and convincing to the target organism. The Rust synthesis.rs module houses the Rosetta Engine, a multi-engine synthesis framework specifically designed to enforce biological realism.¹³

When the Rust layer receives the SynthesisTimeline, it projects the original 112D features into a synthesis-controllable physics intermediary structure encompassing Pitch, Timing, macro-Dynamics, Timbre, advanced Texture (jitter/shimmer/FM depth), and Spatial Control (pan and width).¹³ The system does not rely on a single method of sound generation; rather, it deploys specific, specialized synthesis engines dynamically depending on the required acoustic topology¹³:

Synthesis Engine	Operational Mechanism	Biological Application
Granular Synthesizer	Extracts micro-segments of audio (grains) from high-quality exemplars, applies Hanning window envelopes, and rapidly reconstructs them based on ADSR tracking.	Ideal for replicating complex, highly textured broadband noise and rapid clicks where parametric modeling falls short. ¹³
Dynamic Microharmonic Synthesizer	Calculates an instantaneous fundamental frequency, overlays continuous frequency and amplitude modulation, builds an additive stack of up to eight phase-accumulated harmonics, and injects broadband noise strictly scaled to the target Harmonic-to-Noise Ratio (HNR).	Generates incredibly fluid, natural-sounding tonal phrases, pure whistles, and mammalian sweeps. ¹³

Concatenative Synthesizer	Horizontally combines sequential phonetic elements across a timeline, utilizing mathematically precise 10-millisecond crossfades to blend transitional boundaries.	Executes the sequential playback of complex syntactic sequences, ensuring that the transitions between distinct symbols do not contain digital artifacts.[13, 13]
Superpositional Synthesizer	Manages vertically layered synthesis, allowing multiple generative structures to execute simultaneously.	Absolutely critical for simulating colonial chorusing behaviors, or reproducing the simultaneous multi-frequency calls observed in specific avian syrinx anatomy or marine mammal emissions. ¹³

DDSP Neural Decoder and Jetson Edge Deployment

To achieve unparalleled generative control and biological realism, the most recent update to the Rosetta Engine implements a PyTorch-differentiable DDSP (Differentiable Digital Signal Processing) pipeline.¹³ This pipeline acts as a generative synthesizer that maps the 112D Rosetta Features directly into acoustic waveforms using a DDSPDecoder.¹³ This decoder is a multilayer perceptron (MLP) with a hidden dimension of 256 with dropout regularization that translates the high-dimensional features into 65 specific DDSP parameters: 60 harmonic amplitudes (utilizing softmax) and 5 noise magnitudes (utilizing ReLU).¹³

The acoustic output is rendered through a DifferentiableSineOscillator, which generates phase-continuous sine waves with cumulative phase integration to ensure click-free audio and full gradient tracking, and a DifferentiableNoiseFilter, an FIR filter bank operating across 5 frequency bands using FFT for frequency-domain filtering.¹³

Neural Post-Filter Refinement

To further augment the biological authenticity of the DDSP output—specifically capturing missing micro-features or subtle time-varying harmonic patterns that basic generative parameters might miss—the pipeline incorporates a continuous neural post-filter.¹³ This lightweight Convolutional Neural Network (CNN) comprises approximately 50,000 parameters deployed across four 1D convolutional layers.¹³ The network applies a residual connection architecture where the final acoustic output is computed as the sum of the raw DDSP audio input and the learned CNN refinement, ensuring bounded modifications and stable gradient flow.¹³

During processing, this post-filter is dynamically conditioned on the 65 DDSP parameters (the 60 harmonic amplitudes and 5 noise band magnitudes) via a linear embedding layer.¹³ The model is trained using rigorous augmentations (pitch shifting, time stretching, gain variation, and noise injection) alongside a composite perceptual loss function that strictly combines a Multi-Scale Spectral Loss across multiple resolution frames (512, 1024, and 2048) and a time-domain L1 loss.¹³

To deploy this heavy neural architecture in the field, the system utilizes an ONNX Export (Opset 18) and a TensorRT Builder to apply FP16 optimization specifically for NVIDIA Jetson edge devices.¹³ Managed by the RealtimeDDSPAgent over ZMQ IPC, this architecture is extraordinarily fast. The standalone post-filter boasts a mean inference latency of just 2.9 milliseconds on a Jetson Orin Nano and 4.2 milliseconds on a Jetson Xavier NX.¹³ The total end-to-end synthesis latency, combining the DDSP decoder and the CNN post-filter, achieves 8 milliseconds on the Orin Nano and 12 milliseconds on the Xavier NX, comfortably exceeding the strict sub-50 millisecond real-time biological interaction target.¹³

Species-Specific Synthesis Profiles and Safety Parameters

A universal, generalized synthesizer is biologically impossible; the physics of sound production varies radically across taxa. Consequently, the Rosetta Engine strictly enforces predefined species-specific boundary parameters to ensure the synthesized output is physically possible for the target organism's anatomical constraints.¹³

For example, the operational profile for the common marmoset (*Callithrix jacchus*) mathematically constrains the engine to an F0 range of 6 to 12 kHz, centering around a mean of 8 kHz. Synthesized grains are mandated to endure for 50 to 80 milliseconds, accented by a moderate, biologically accurate vibrato of 6 to 8 Hz.¹³ Conversely, the profile for the Egyptian fruit bat (*Rousettus aegyptiacus*) demands an entirely different, extreme acoustic domain. The bat synthesis engine must execute at massive sample rates (e.g., 96 kHz) to prevent digital aliasing of ultrasonic output.¹³ Bat fundamentals are scaled dynamically between 20 and 100 kHz, necessitating vastly shorter grain durations (20 to 50 milliseconds) to accommodate ultra-fast vibrato rates (8 to 12 Hz) and massive, rapid frequency modulations spanning up to 10,000 Hz in a single sweep.¹³

To guarantee both system stability and animal welfare, the synthesis engine features an `emergency_stop()` interrupt function, which can instantly clear all active grains and wipe the output buffers to zero.¹³ Furthermore, automatic scaling and sample normalization are hardcoded into the output pipeline to physically prevent audio clipping and the resulting sudden, potentially harmful spikes in acoustic pressure.¹³

Empirical Findings: Unlocking the Bat Corpus (*Rousettus aegyptiacus*)

The robustness and immense analytical power of the intra-call acoustic-first pipeline have been

rigorously validated through extensive empirical applications in highly social model organisms. The Egyptian fruit bat (*Rousettus aegyptiacus*) serves as an ideal biological model for evaluating ALP due to its incredibly complex colonial structures, near-constant social interactions, and expansive, high-frequency vocal repertoire.³⁸

Historically, human observers and traditional ethologists dismissed the chaotic cacophony of a bat colony as generic, non-specific aggressive noise, interpreting nearly all vocalizations as a simple, uniform signal indicating distress or annoyance.⁴⁰ However, by deploying advanced machine learning algorithms conceptually identical to the acoustic-first pipeline over a continuous, 75-day observational dataset comprising over 15,000 individually recorded calls, researchers at Tel Aviv University demonstrated profound, previously unimaginable semantic specificity.³⁹

The continuous feature extraction confirmed that the bat noises are not random variations. In fact, the models successfully classified over 60 percent of the recorded vocalizations into four highly distinct, nuanced situational categories.⁴⁰ Rather than emitting a generic alarm, the bats actively encoded specific situational data within the intra-call continuous variables to explicitly convey whether a social dispute involved the competitive allocation of food, spatial positioning conflicts within a densely packed sleeping cluster, the rejection of unwanted mating advances by males, or simple proximity violations by neighboring individuals.⁴⁰

Furthermore, the extracted multidimensional features confirmed a sophisticated level of social addressability—directly fulfilling Level 3 Data requirements (Interaction-Ready). The analysis revealed that bats alter their precise vocal parameters depending on the specific individual they are addressing within the colony.⁴⁰ This behavior mirrors human prosodic shifts—using a different tone of voice when speaking to different people—and underscores the absolute necessity of continuous, intra-call acoustic analysis, as binary CPD segmentation would fundamentally erase these critical micro-variations.

Crowd-Based Vocal Learning and the Acquisition of Dialects

Beyond static context mapping, the deployment of machine learning in bioacoustics has been deeply instrumental in addressing one of the most profound mysteries in biology: the phylogenetic origins of language acquisition.³⁸ Vocal learning—the advanced neurological ability to actively modify acoustic production based on auditory experience and social feedback—is the foundational prerequisite for human language development.³⁸ While heavily documented and extensively studied in songbirds, compelling evidence of complex vocal learning in non-human mammals has historically been sparse and contested.³⁸

Controlled isolation and interactive playback experiments tracking the ontogeny of bat calls continuously from birth to adulthood have definitively demonstrated sophisticated vocal learning mechanisms in *Rousettus aegyptiacus*.³⁸ In strict isolation experiments, bat pups raised without acoustic access to the broader colony exhibited severely underdeveloped, anomalous vocal repertoires.³⁸ Conversely, in experiments where isolated pups were exposed to artificial

playbacks manipulating the fundamental frequency of the typical colony call, the developing bats actively shifted their own physiological F0 usage to mathematically match the artificial playback.⁴²

These findings present a paradigm shift in our understanding of mammalian learning. They suggest that, unlike the direct, one-to-one parent-to-offspring transmission typically observed in certain avian models, bat vocal acquisition is heavily reliant on a distributed "crowd dialect".³⁹ Pups learn to vocalize by observing, processing, and absorbing the statistical distribution of acoustic features from the surrounding colony of hundreds of individuals, rather than solely mimicking their mother's specific "tongue".³⁹ The researchers likened this to a human child adopting a regional dialect, such as a London or Scottish accent, from their local environment rather than strictly inheriting the accent of their parents.³⁹ The emergence of dialect variation based on macro-social exposure heavily reinforces the requirement for Level 2 (Functional) data profiling; a generic, species-wide synthesis model would fail to match the specific, mathematically distinct dialectal properties of an isolated colony, rendering closed-loop playback experiments ineffective or confusing to the animals.¹³

The Emergence of Combinatorial Syntax

As machine learning systems possess the processing power to uncover the deep depth of intra-call variations, they simultaneously reveal highly complex structural rules governing inter-call relationships. Advanced syntactic mining of these acoustic features has successfully isolated the presence of robust combinatorial syntax in several mammalian and avian species.¹⁸ In combinatorial syntactic structures, discrete acoustic units—which may carry little to no semantic meaning in total isolation—are sequentially linked, arranged, and combined in varying hierarchies to generate complex, novel meanings.²³

While humans maintain a monopoly on complex phonological syntax (where the precise arrangement of phonemes fundamentally alters lexical meaning, creating distinct words)¹⁸, species such as bats, cetaceans, elephants, and specific primates clearly utilize usage-based combinatorial sequencing to dynamically negotiate social boundaries.⁵ Research into free-tailed bats has empirically confirmed that song syntax is not a rigid, biologically hardcoded output; rather, it undergoes rapid, highly complex sequential alterations dependent entirely on shifting social and environmental contexts.⁵ An artificial closed-loop interaction agent must therefore possess the computational capability not merely to identify a static call type, but to fully comprehend the latent temporal rules governing its syntactic construction, actively concatenating acoustic exemplars dynamically to reflect an ongoing, fluid behavioral shift in real time.[5, 13, 13]

Field Deployment and Ethological Validation Protocols

To formally evaluate the capability of the InteractionAgent (currently at v1.5.0) in achieving the Turing-level biological threshold, the methodology dictates a rigorous Ethological Validation Protocol (v1.0.0).¹³ This framework explicitly measures "Syntactic Continuity"—the premise that

an animal will only accept a synthetic response if it is contextually appropriate, utilizes correct combinatorial syntax (e.g., executing only the 50 valid bigrams within the bat matrix), and sounds acoustically authentic, relying heavily on exemplar-based synthesis rather than fully generative approximations.¹³

The primary quantitative metric for evaluating this biological acceptance is the Response Appropriateness Score (RAS).¹³ The system measures whether an interacting animal continues a syntactic chain following a machine-generated response within a specific timeout window (e.g., 2 seconds).¹³ It is calculated via the specific formula:

$$R = \frac{\text{Number of valid follow-up responses}}{\text{Total system responses}}$$

The resulting metric strictly dictates functional acceptance: an $R \geq 0.7$ indicates full acceptance (meaning the system is actively participating as a conspecific), scores between 0.5 and 0.7 indicate partial acceptance, and an $R < 0.3$ constitutes complete rejection (the system is detected as artificial).¹³

To execute these measurements, field deployments utilize an Observation Zone equipped with a directional speaker and a microphone array.¹³ The validation protocol cycles through an acclimation period and a passive baseline, eventually testing five distinct conditions:

1. **Baseline (Control):** Passive recording with zero system intervention.
2. **Conspecific Playback:** A positive control utilizing standard pre-recorded bat calls.
3. **Full System:** Active closed-loop interaction utilizing mathematically valid bigrams and syntax.
4. **Invalid Syntax:** A negative control forcing random bigram transitions to explicitly test whether the subject animal will reject incorrect syntactic chains.
5. **Synthetic Tones:** An acoustic control utilizing pure continuous tones to test acoustic rejection.¹³

Statistical success under this protocol mandates that the Full System achieves an $R \geq 0.7$ alongside a significant increase in RAS compared to the baseline ($p < 0.05$), while simultaneously showing no significant difference when compared against the Conspecific Playback ($p > 0.05$).¹³ Crucially, the Invalid Syntax condition must yield an $R < 0.3$, definitively proving that the interacting organism is sensitive to the artificial syntax generated.¹³

Finally, extreme safety constraints are integrated directly into the field protocol to preserve animal welfare. Active playback interactions are limited to two hours daily and are hardware-capped at an output volume of 85 dB SPL at 1 meter.¹³ The testing sequence will automatically abort if the NBD pipeline detects that colony alarm vocalizations have spiked by

300% over the recorded baseline, and strict watchdog timers enforce an immediate system mute if the software logic layer drops communication for over 100 milliseconds.¹³ Core components of this validation tracking have already been successfully deployed and validated across 22 individual field tests.¹³

Future Directions: Multimodality, Ethics, and the Interspecies Turing Test

The successful integration of $\mathcal{O}(n)$ Neural Boundary Detection, high-dimensional 112D Rosetta feature extraction, unsupervised symbol clustering, and ultra-low-latency bidirectional interaction protocols provides a comprehensive, scalable, and scientifically rigorous solution to the two-way interspecies communication problem. By operating directly upon the continuous acoustic physics of the biological call, the methodology systematically strips away the anthropocentric biases that have historically constrained the study of animal linguistics.¹⁷ It empowers computational systems to detect both the rigid syntactical architecture and the highly fluid, affective intra-call continuous gradients that dictate actual biological meaning.⁴

The overwhelming success of these AI-driven mechanisms has rapidly propelled the field of digital bioacoustics toward a momentous, historic threshold: the realization of a biological Interspecies Turing Test.¹⁵ Drawing directly from Alan Turing's original philosophical arguments regarding behavioral inference and the necessity of modeling the internal state of "the other" through observable interactions⁴⁴, initiatives such as the Collier-Dolittle Challenge are presently offering substantial, multi-million dollar financial incentives to accelerate this breakthrough.¹² A Grand Prize of ten million dollars will be awarded to researchers capable of deploying artificial algorithms that can successfully interface with non-human organisms in an autonomous, contextually accurate, and entirely non-invasive manner.¹² To formally satisfy the parameters of this Turing test, the artificial closed-loop agent must actively participate in a sustained two-way dialogue across multiple, distinct behavioral contexts (e.g., foraging negotiation, alarm broadcasting, mating rituals) using the organism's endogenous signals, all while maintaining such biological fidelity that the subject animal acts entirely independently, utterly failing to recognize that it is communicating with a synthetic computational entity rather than a conspecific.¹⁵

However, achieving this ultimate objective requires the continued evolution and refinement of foundation models and the expansion of the current acoustic-only paradigm into comprehensive multimodality.⁸ Animals do not interact within an acoustic vacuum; their communication is inherently multimodal, inextricably linked to physical movement, visual displays, tactile interaction, and environmental positioning.⁸ While the current intra-call acoustic-first framework is highly proficient, future iterations must systematically integrate correlative data streams to capture the full biological reality.⁸ Sources of this multimodal data will include animal-borne video bio-loggers, networked camera traps, and advanced visual kinematics models (such as DeepLabCut) capable of tracking structural changes in the animal's physical posture

synchronized with the acoustic output.⁸

Finally, as the computational capability to seamlessly encode, decode, and actively speak the vernacular of wild species matures from a theoretical possibility to an operational reality, profound ethical and philosophical obligations arise.¹ The Earth Species Project, in partnership with The Collective Intelligence Project, recently conducted a first-of-its-kind global survey across 67 countries, polling over a thousand individuals regarding the ethics of AI-powered interspecies communication.⁴⁶ The findings revealed a complex global public deeply curious about the inner lives of other species, yet highly apprehensive regarding the profound risks.⁴⁶

The core ethical dilemma is stark: the exact same foundational AI models and closed-loop technologies that enable conservation scientists to monitor biodiversity health, understand ecological stressors, and potentially prevent species collapse could theoretically be repurposed for mass exploitation.¹ If access to interspecies translation protocols becomes unregulated, these tools could easily be weaponized for the industrial manipulation of wildlife, increasing the efficiency of commercial fishing, enabling highly targeted illicit poaching, or allowing for the unnatural domestication and behavioral control of wild populations for tourism.¹ The deployment of closed-loop interaction agents must therefore be strictly governed by robust, internationally recognized bioethical frameworks that mandate non-invasive interaction protocols and legally prioritize the preservation of natural behavioral hierarchies above human curiosity.¹

The convergence of deep computational acoustics, real-time edge processing, large language models, and ethological science represents a definitive, permanent leap forward in the Earth sciences. By discarding human linguistic bias and listening directly to the fundamental physical parameters of the natural world, the capability to understand—and eventually converse dynamically with—the vast, diverse expanse of planetary intelligence is no longer a work of science fiction, but an imminent, rigorously defined computational reality.

Works cited

1. Animal Language Processing: An AI Convergence In Animal Communication, accessed April 27, 2026, <https://earthspecies.org/2026/04/02/animal-language-processing-an-ai-convergence-in-animal-communication/>
2. Computational bioacoustics with deep learning: a review and roadmap - PMC, accessed April 27, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC8944344/>
3. AI Advances Enable Prospects for Animal Communication | Let's Data Science, accessed April 27, 2026, <https://letsdatascience.com/news/ai-advances-enable-prospects-for-animal-communication-f57bbf1c>
4. ORCA-SLANG: An Automatic Multi-Stage Semi-Supervised Deep Learning Framework for Large-Scale Killer Whale Call Type Identification - ISCA Archive, accessed April 27, 2026, https://www.isca-archive.org/interspeech_2021/bergler21_interspeech.pdf
5. Social context evokes rapid changes in bat song syntax - ResearchGate,

- accessed April 27, 2026,
https://www.researchgate.net/publication/243826998_Social_context_evokes_rapid_changes_in_bat_song_syntax
6. Community Bioacoustics: Studying Acoustic Community Structure for Ecological and Conservation Insights - Frontiers, accessed April 27, 2026,
<https://www.frontiersin.org/journals/ecology-and-evolution/articles/10.3389/fevo.2021.706445/full>
 7. Digital Bioacoustics: Will AI Allow People To Talk With Their Pets In The Future?, accessed April 27, 2026,
<https://www.hybridvigoremory.org/articles/digital-bioacoustics-will-ai-allow-people-to-talk-with-their-pets-in-the-future>
 8. ESP Technical Roadmap - Earth Species Project, accessed April 27, 2026,
<https://earthspecies.org/2022/12/02/esp-technical-roadmap/>
 9. Artificial Intelligence and the Future of Interspecies Communication - Houston Foresight, accessed April 27, 2026,
<https://www.houstonforesight.org/artificial-intelligence-and-the-future-of-interspecies-communication/>
 10. What We Do - Earth Species Project, accessed April 27, 2026,
<https://earthspecies.org/what-we-do/>
 11. Earth Species Project – More than 8 million species share our planet. We only understand the language of one., accessed April 27, 2026,
<https://earthspecies.org/>
 12. \$10 million prize for Interspecies Two-Way Communication using generative AI models announced - Jeremy Collier Foundation, accessed April 27, 2026,
<https://www.jeremycolliderfoundation.org/news-and-insights/press-releases/10-million-prize-for-interspecies-two-way-communication-using-generative-ai-models-announced>
 13. closed_loop_agent_protocol.md
 14. A theory of how active behavior stabilises neural activity: Neural gain modulation by closed-loop environmental feedback - PMC, accessed April 27, 2026,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC5809098/>
 15. Collier-Dolittle Prize, accessed April 27, 2026,
<https://coller-dolittle-24.sites.tau.ac.il/>
 16. Dolphin whistle decoders win \$100000 interspecies communication prize - The Guardian, accessed April 27, 2026,
<https://www.theguardian.com/world/2025/may/15/us-scientists-who-deciphered-dolphin-whistles-win-prize-for-animal-communication>
 17. Journal of Speech, Language, and Hearing Research | Vol 69, No 4, accessed April 27, 2026, <https://pubs.asha.org/toc/jslhr/current>
 18. Language evolution: syntax before phonology? - PMC - NIH, accessed April 27, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC4083781/>
 19. The Development of Speech Processing Strategies for the University of Melbourne/Cochlear Multiple Channel Implantable Hearing Prosthesis, accessed April 27, 2026,
https://cjslpa.ca/files/1992_JSLPA_Vol_16/No_02_89-176/Clark_JSLPA_1992.pdf

20. Scopus+Web of Science all var sounds - PeerJ, accessed April 27, 2026,
https://peerj.com/preprints/6570v2/Supplemental_1.xlsx
21. Acoustic Dictionary - explanation of key concepts and physical laws - Nyquista, accessed April 27, 2026,
<https://nyquista.pl/en/blog/categories/dictionary-acoustics/>
22. Untitled - eDiss - Universität Göttingen, accessed April 27, 2026,
https://ediss.uni-goettingen.de/bitstream/11858/00-1735-0000-0001-BB74-0/1/Tabitha%20Price_PhD%20thesis_ediss.pdf
23. Syntactic properties of male courtship song in the lesser short-tailed bat, *Mystacina tuberculata* - Frontiers, accessed April 27, 2026,
<https://www.frontiersin.org/journals/ecology-and-evolution/articles/10.3389/fevo.2022.907791/full>
24. Kinematic and Acoustic Responses to Predictable and Unexpected Auditory Feedback Perturbations in Speakers With and Without Parkinson's Disease | Request PDF - ResearchGate, accessed April 27, 2026,
https://www.researchgate.net/publication/402790190_Kinematic_and_Acoustic_Responses_to_Predictable_and_Unexpected_Auditory_Feedback_Perturbations_in_Speakers_With_and_Without_Parkinson's_Disease
25. Day 1 - 16th April 2026 - gasla-18 - University of Cambridge, accessed April 27, 2026, <https://gasla18.mml.cam.ac.uk/assets/GASLA18-PROGRAM.pdf>
26. No Effect of Note Order on the Response of Coal Tits to Conspecific, Heterospecific and Artificial Mobbing Calls - Imperial Spiral, accessed April 27, 2026,
<https://spiral.imperial.ac.uk/server/api/core/bitstreams/174fb7ae-b24b-49fe-ba24-32cc146378d2/content>
27. Call patterns encode and transmit emotion in marmoset monkeys - bioRxiv, accessed April 27, 2026,
<https://www.biorxiv.org/content/10.1101/2022.08.03.502601.full>
28. Complex rules of vocal sequencing in marmoset monkeys - bioRxiv, accessed April 27, 2026, <https://www.biorxiv.org/content/10.1101/2022.08.03.502601v1.full>
29. Social context evokes rapid changes in bat song syntax, accessed April 27, 2026,
[https://engdes.com/sigwin/company/biblio/papers/Social%20context%20evokes%20rapid%20changes%20in%20bat%20song%20syntax%20\(Bohn%20et%20al%202013\).pdf](https://engdes.com/sigwin/company/biblio/papers/Social%20context%20evokes%20rapid%20changes%20in%20bat%20song%20syntax%20(Bohn%20et%20al%202013).pdf)
30. Decoupled Acoustic and Visual Components in the Multimodal Signals of the Common Cuckoo (*Cuculus canorus*) - Frontiers, accessed April 27, 2026,
<https://www.frontiersin.org/journals/ecology-and-evolution/articles/10.3389/fevo.2021.725858/full>
31. Form and Function in Vocal Communication (Chapter 3) - Cambridge University Press, accessed April 27, 2026,
<https://www.cambridge.org/core/books/animal-vocal-communication/form-and-function-in-vocal-communication/FC07D0A2A1E494FB9D70BDAB2EE411B9>
32. HybridMouse: A Hybrid Convolutional-Recurrent Neural Network-Based Model for Identification of Mouse Ultrasonic Vocalizations - Frontiers, accessed April 27, 2026,

- <https://www.frontiersin.org/journals/behavioral-neuroscience/articles/10.3389/fnbeh.2021.810590/full>
33. Automated annotation of birdsong with a neural network that segments spectrograms - eLife, accessed April 27, 2026, <https://elifesciences.org/articles/63853>
 34. Closed-Loop Control of Active Sensing Movements Regulates Sensory Slip | bioRxiv, accessed April 27, 2026, <https://www.biorxiv.org/content/10.1101/366609v1.full-text>
 35. Automatic segmentation and classification of mice ultrasonic vocalizations - AIP Publishing, accessed April 27, 2026, <https://pubs.aip.org/asa/jasa/article/152/1/266/2838328/Automatic-segmentation-and-classification-of-mice>
 36. DeepSqueak: a deep learning-based system for detection and analysis of ultrasonic vocalizations - PMC, accessed April 27, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC6461910/>
 37. Closed-Loop Control of Active Sensing Movements Regulates Sensory Slip† - Johns Hopkins University, accessed April 27, 2026, <https://limbs.lcsr.jhu.edu/wp-content/papercite-data/pdf/biswasclosed-loop2018.pdf>
 38. Vocal learning in a social mammal: Demonstrated by isolation and playback experiments in bats - PMC, accessed April 27, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC4643821/>
 39. Language school for bats | Tel Aviv University, accessed April 27, 2026, https://english.tau.ac.il/news/bats_language_school
 40. Researchers 'Translate' Bat Talk. Turns Out, They Argue—a Lot - Smithsonian Magazine, accessed April 27, 2026, <https://www.smithsonianmag.com/smart-news/researchers-translate-bat-talk-and-they-argue-lot-180961564/>
 41. The multi-dimensional nature of vocal learning - Royal Society Publishing, accessed April 27, 2026, <https://royalsocietypublishing.org/rstb/article/376/1836/20200236/108664/The-multi-dimensional-nature-of-vocal-learningThe>
 42. Crowd vocal learning induces vocal dialects in bats: Playback of conspecifics shapes fundamental frequency usage by pups | PLOS Biology - Research journals, accessed April 27, 2026, <https://journals.plos.org/plosbiology/article?id=10.1371/journal.pbio.2002556>
 43. Scientists have almost cracked the secret language of animals. Here's what they've learned, accessed April 27, 2026, <https://www.sciencefocus.com/future-technology/secret-language-animals-ai>
 44. Transcript 7 | Can Machines Learn How to Behave? with Blaise Agüera y Arcas, accessed April 27, 2026, <https://www.interspecies.io/transcripts/blaise>
 45. Real-Time Closed-Loop Feedback in Behavioral Time Scales Using DeepLabCut | eNeuro, accessed April 27, 2026, <https://www.eneuro.org/content/8/2/ENEURO.0415-20.2021>
 46. What the World Thinks About AI and Animal Communication: Findings from Our

First Global Survey - Earth Species Project, accessed April 27, 2026,
<https://earthspecies.org/2025/09/29/what-the-world-thinks-about-ai-and-animal-communication-findings-from-our-first-global-survey/>